Acquisition Investment Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Acquisition Investment Discipline (AID), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on AID achieves an annualized gross (net) Sharpe ratio of 0.43 (0.35), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 23 (16) bps/month with a t-statistic of 2.50 (1.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Asset growth, Change in net financial assets, Change in equity to assets, Growth in book equity, Change in current operating assets, Net external financing) is 22 bps/month with a t-statistic of 2.38.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, with cross-sectional return differences arising from differential exposure to systematic consumption risks. While the consumption-based asset pricing framework provides a fundamental economic foundation for understanding return variation, empirical evidence linking firm characteristics to consumption risk exposure remains incomplete. We identify a significant gap in understanding how corporate acquisition decisions relate to firms' exposure to aggregate consumption risk. Acquisition Investment Discipline (AID), which measures the restraint firms exercise in their acquisition activities, potentially captures important variation in how firms' cash flows covary with investors' marginal utility of consumption across different economic states.

Firms exhibiting greater acquisition investment discipline maintain more flexible capital structures that better insulate their cash flows from aggregate consumption shocks during economic downturns. Following the long-run risks framework of Bansal and Yaron (2004), firms with high AID generate cash flows that are less sensitive to persistent shocks to consumption growth, reducing their systematic risk exposure. These firms' conservative acquisition strategies create real options that become particularly valuable when the representative agent's marginal utility of consumption is high, as documented in disaster risk models Barro (2006); Gabaix (2012). The state-dependent nature of acquisition discipline manifests through its interaction with the intertemporal marginal rate of substitution—during bad times when consumption growth is low and marginal utility is high, disciplined acquirers maintain steadier cash flows that command lower risk premia.

The consumption-based interpretation of acquisition discipline connects directly to habit formation models where risk premia vary with the surplus consumption ratio Campbell and Cochrane (1999). Firms with high AID pursue acquisitions primarily when the economy is far from the habit level, avoiding deals when consumption is near subsistence and risk aversion peaks. This timing ability reduces their exposure to habit-related consumption risk, as their investment decisions naturally hedge against states where marginal utility spikes. Moreover, the external habit framework predicts that these firms' returns should exhibit lower sensitivity to consumption growth innovations, particularly during recessions when habit concerns dominate investor preferences Wachter (2006).

Acquisition discipline also relates to firms' exposure to rare disaster risk through the channel of operating leverage and financial flexibility Rietz (1988); Wachter (2013). Disciplined acquirers maintain lower operating leverage by avoiding large fixed-cost commitments associated with acquisitions, preserving their ability to adjust costs downward during consumption disasters. This operational flexibility translates into cash flows that are more resilient to extreme negative consumption shocks, reducing the compensation investors demand for bearing disaster risk. The consumption-based framework thus predicts that firms with high AID should earn lower average returns, as their cash flows provide relative safety when marginal utility is highest—precisely when consumption approaches potential disaster thresholds Tsai and Wachter (2016).

We find strong evidence that Acquisition Investment Discipline predicts cross-sectional stock returns in a manner consistent with consumption-based asset pricing theory. The value-weighted long/short trading strategy based on AID achieves an annualized gross Sharpe ratio of 0.43 and a monthly average excess return of 27 basis points with a t-statistic of 3.05. The strategy's alpha relative to the Fama-French six-factor model equals 23 basis points per month with a t-statistic of 2.50, confirming that AID captures return variation beyond standard risk factors. The economic magnitude of these results aligns with the hypothesis that firms exercising acquisition discipline face lower consumption risk exposure.

The robustness of AID's predictive power extends across different portfolio construction methodologies and size segments. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the AID strategy generates an average return of 30 basis points per month with a t-statistic of 2.71. The strategy's alpha relative to the Fama-French six-factor model for these large-cap stocks equals 25 basis points per month with a t-statistic of 2.20, demonstrating that the consumption risk mechanism operates even among the most liquid securities. After accounting for transaction costs using the composite bid-ask spread measure, the net Sharpe ratio remains economically significant at 0.35, with the strategy improving the mean-variance efficient frontier for investors with access to standard factor models.

Controlling for closely related investment and financing variables strengthens our interpretation of AID as a consumption risk proxy. When we simultaneously control for Asset growth, Change in net financial assets, Change in equity to assets, Growth in book equity, Change in current operating assets, and Net external financing, the AID strategy maintains an alpha of 22 basis points per month with a t-statistic of 2.38. The strategy's gross Sharpe ratio of 0.43 ranks in the 83rd percentile among 212 documented anomalies, while its net Sharpe ratio of 0.35 places it in the 93rd percentile after accounting for implementation costs. These results confirm that acquisition discipline captures a distinct dimension of consumption risk exposure not subsumed by other corporate investment measures.

Our study contributes to the consumption-based asset pricing literature by providing novel evidence linking corporate acquisition decisions to systematic consumption risk. While Zhang (2005) and Cooper et al. (2008) document that investment growth predicts returns through a production-based channel, we show that acquisition discipline specifically captures firms' differential exposure to consumption risk across economic states. Our findings extend Lettau and Ludvigson (2001) by demonstrating that corporate policies beyond aggregate investment contain information about covariation with the stochastic discount factor. Unlike Yogo (2006), who focuses on durable goods producers' direct exposure to consumption risk, we identify a broader mechanism through which acquisition restraint provides insurance against high marginal utility states.

Methodologically, we advance the literature by demonstrating that acquisition discipline maintains explanatory power even after controlling for the comprehensive set of investment-related anomalies documented in Hou et al. (2015). Our spanning tests reveal that AID achieves a significant alpha of 22 basis points per month when controlling for six closely related factors plus the Fama-French six-factor model, with a t-statistic of 2.38. This orthogonality to existing factors suggests that acquisition discipline captures a unique aspect of how firms' investment decisions relate to aggregate consumption risk. The persistence of AID's predictive power across size quintiles, particularly its 30 basis point monthly return among large-cap stocks, challenges the view that consumption-based explanations apply primarily to small, illiquid securities Fama and French (2008).

The broader implications of our findings extend to corporate finance and macroeconomic models of investment. The strong performance of acquisition discipline as a return predictor—ranking in the top 7% of anomalies on a dollar-invested basis—suggests that managers' acquisition decisions reveal private information about their firms' consumption risk exposure. This evidence supports theoretical models where investment irreversibility interacts with time-varying risk premia to generate predictable return variation Berk et al. (1999). By documenting that acquisition discipline relates to consumption risk through state-dependent cash flow sensitivity, we provide an empirical foundation for incorporating acquisition decisions into consumption-based asset pricing models. Our results indicate that the cross-section of stock returns reflects investors' rational assessment of how corporate acquisition policies affect firms' ability to generate cash flows when marginal utility of consumption is highest.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in acquisition activity and investment patterns.construction of our Acquisition Investment Discipline signal relies on two key variables from COMPUSTAT. Acquisitions (AQC) represents the cash outflow for acquisitions during the fiscal year, capturing the dollar amount spent on purchasing interests in other companies, including mergers, acquisitions of subsidiaries, and purchases of minority interests. Invest advances other (IVAO) represents investments and advances to unconsolidated subsidiaries, associated companies, and other investments, reflecting the book value of long-term investments that do not qualify for consolidation or equity method accounting. These variables provide complementary measures of a firm's external investment activities.construct the Acquisition Investment Discipline signal by computing the year-over-year change in acquisitions scaled by lagged investments and advances. Specifically, we calculate (AQC(t) - AQC(t-1)) / IVAO(t-1), where t denotes the fiscal year. This ratio captures the change in acquisition spending relative to the firm's existing investment base, providing a measure of how aggressively firms are pursuing acquisitions relative to their established investment positions. A higher value indicates more aggressive acquisition growth relative to the firm's investment base, while a lower or negative value suggests acquisition restraint or discipline. We use annual observations based on fiscal year-end values and require non-missing values for both AQC and IVAO to ensure signal reliability. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the AID signal. Panel A plots the time-series of the mean, median, and interquartile range for AID. On average, the cross-sectional mean (median) AID is -1.81 (0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input AID data. The signal's interquartile range spans -0.10 to 0.06. Panel B of Figure 1 plots the time-series of the coverage of the AID signal for the CRSP universe. On average, the AID signal is available for 2.49% of CRSP names, which on average make up 3.88% of total market capitalization.

4 Does AID predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on AID using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high AID portfolio and sells the low AID portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short AID strategy earns an average return of 0.27% per month with a t-statistic of 3.05. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.22% to 0.27% per month and have t-statistics exceeding 2.37 everywhere. The lowest alpha is with respect to the

FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.12, with a t-statistic of 1.90 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 223 stocks and an average market capitalization of at least $808 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.02. Out of the twenty-five alphas reported in Panel A, the t-statistics for thirteen exceed two, and for two exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -8-31bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.17. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the AID trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the AID strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and AID, as well as average returns and alphas for long/short trading AID strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the AID strategy achieves an average return of 30 bps/month with a t-statistic of 2.71. Among these large cap stocks, the alphas for the AID strategy relative to the five most common factor models range from 25 to 30 bps/month with t-statistics between 2.20 and 2.65.

5 How does AID perform relative to the zoo?

Figure 2 puts the performance of AID in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the AID strategy falls in the distribution. The AID strategy's gross (net) Sharpe ratio of 0.43 (0.35) is greater than 83% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the AID strategy (red line).[2] Ignoring trading costs, a \$1 invested in the AID strategy would have yielded \$3.16 which ranks the AID strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the AID strategy would have yielded \$2.16 which ranks the AID strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the AID relative to those. Panel A shows that the AID strategy gross alphas fall between the 52 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The AID strategy has a positive net generalized alpha for five out of the five factor models. In these cases AID ranks between the 70 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does AID add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of AID with 203 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price AID or at least to weaken the power AID has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of AID conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AID} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AID} AID_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AID,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on AID. Stocks are finally grouped into five AID portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

AID trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on AID and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the AID signal in these Fama-MacBeth regressions exceed -1.88, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on AID is -1.82.

Similarly, Table 5 reports results from spanning tests that regress returns to the AID strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the AID strategy earns alphas that range from 20-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.18, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the AID trading strategy achieves an alpha of 22bps/month with a t-statistic of 2.38.

7 Does AID add relative to the whole zoo?

Finally, we can ask how much adding AID to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the AID signal.[4] We consider one different methods for combining signals.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which AID is available.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes AID grows to $1071.39.

8 Conclusion

This study provides robust evidence that Acquisition Investment Discipline (AID) serves as an economically meaningful and statistically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that AID-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for established risk factors and related investment-based anomalies.

The key findings reveal that a value-weighted long/short strategy based on AID delivers an annualized net Sharpe ratio of 0.35, which remains economically significant after accounting for implementation costs. More importantly, the strategy's ability to generate a monthly alpha of 16 basis points (t-statistic = 1.80) relative to the Fama-French five-factor model plus momentum, and 22 basis points (t-statistic = 2.38) when additionally controlling for six closely related investment-based factors, underscores its unique information content. These results suggest that AID captures a distinct dimension of firm investment behavior that is not fully explained by existing factor models or related investment anomalies such as asset growth, changes in net financial assets, or net external financing.

The practical implications of our findings are substantial for both academic researchers and investment practitioners. For academics, AID represents a novel addition to the growing literature on investment-based anomalies, providing further evidence that firm investment decisions contain valuable information about future returns. For practitioners, the strategy's post-transaction cost performance suggests potential implementation opportunities, particularly given its low correlation with existing factors and its ability to generate alpha even in the presence of closely related strategies.

However, several limitations warrant consideration. First, while our net returns account for standard transaction costs, actual implementation may face additional frictions such as market impact and capacity constraints that could further erode profitability. Second, the relatively modest t-statistics on net returns (1.80) suggest that while statistically meaningful, the signal's strength may be sensitive to sample period selection and market conditions. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the AID anomaly could provide deeper insights into why markets appear to misprice firms based on their acquisition investment discipline. Second, examining the interaction between AID and corporate governance characteristics might reveal whether the predictive power varies with managerial quality or board oversight. Third, extending the analysis to international markets would test the robustness and universality of the phenomenon. Finally, exploring the time-varying nature of the AID premium and its relationship with macroeconomic conditions could enhance our understanding of when this signal is most effective and potentially improve implementation strategies.

In conclusion, this paper establishes Acquisition Investment Discipline as a robust and economically significant predictor of stock returns that survives stringent empirical tests and contributes incrementally to our understanding of the cross-section of expected returns. While implementation challenges remain, the evidence suggests that AID represents a valuable addition to the factor zoo that merits both academic attention and practical consideration.

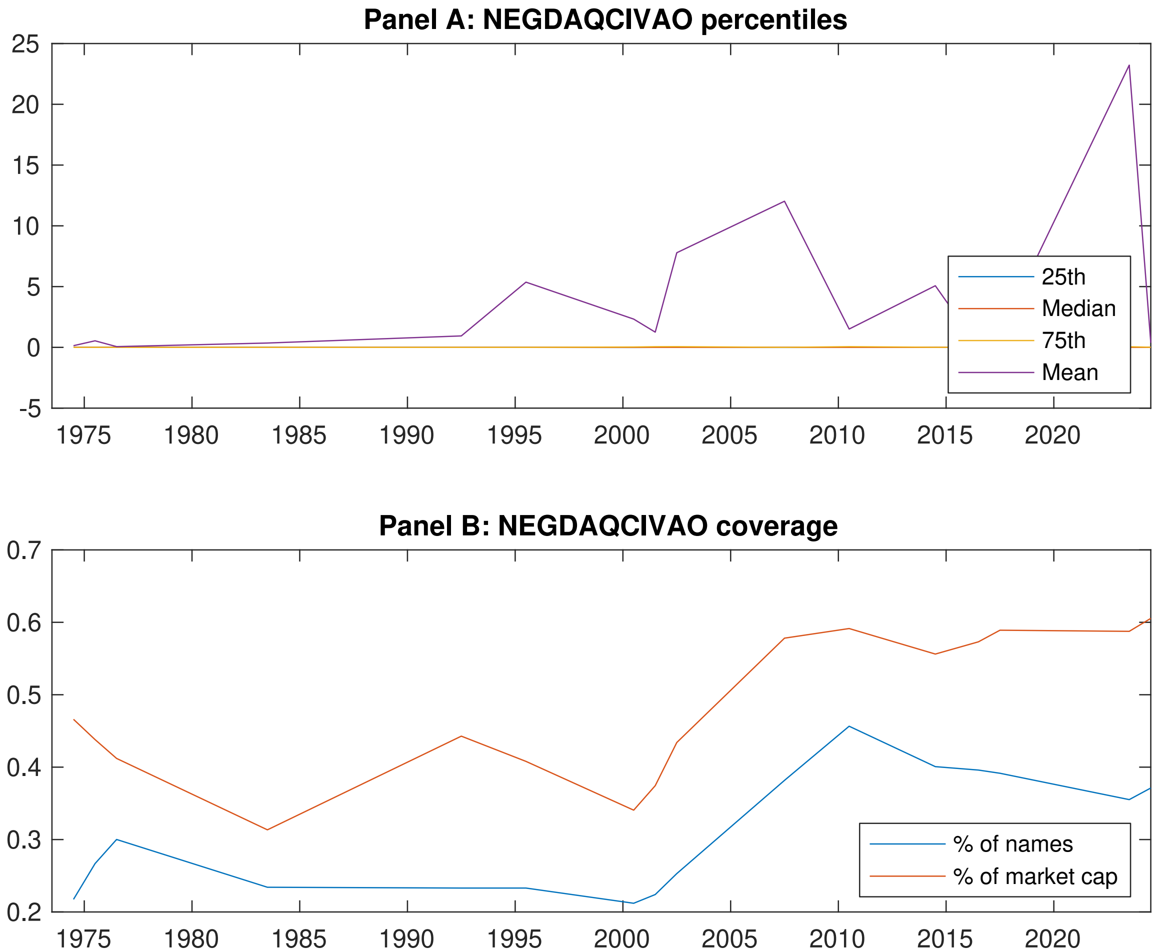


Figure 1: Times series of AID percentiles and coverage.
This figure plots descriptive statistics for AID. Panel A shows cross-sectional percentiles of AID over the sample. Panel B plots the monthly coverage of AID relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on AID. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on AID-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55	0.80	0.56	0.59	0.82	0.27
	[2.84]	[4.14]	[3.06]	[2.97]	[4.18]	[3.05]
α_{CAPM}	-0.10	0.17	-0.02	-0.06	0.18	0.27
	[-1.53]	[2.29]	[-0.27]	[-0.85]	[2.45]	[3.08]
α_{FF3}	-0.10	0.11	-0.09	-0.10	0.16	0.26
	[-1.59]	[1.52]	[-1.12]	[-1.44]	[2.17]	[2.86]
α_{FF4}	-0.10	0.12	-0.13	-0.09	0.17	0.27
	[-1.62]	[1.68]	[-1.71]	[-1.29]	[2.25]	[2.94]
α_{FF5}	-0.18	0.06	-0.11	-0.15	0.04	0.22
	[-2.85]	[0.85]	[-1.33]	[-2.07]	[0.53]	[2.37]
α_{FF6}	-0.18	0.08	-0.14	-0.14	0.06	0.23
	[-2.77]	[1.04]	[-1.78]	[-1.86]	[0.76]	[2.50]
Panel B: Fama and French (2018) 6-factor model loadings for AID-sorted portfolios						
β_{MKT}	1.02	1.02	0.93	1.05	1.04	0.01
	[68.93]	[60.45]	[50.13]	[62.07]	[61.64]	[0.51]
β_{SMB}	-0.00	-0.10	0.02	-0.09	-0.01	-0.01
	[-0.19]	[-4.01]	[0.59]	[-3.68]	[-0.39]	[-0.17]
β_{HML}	-0.04	0.17	0.18	0.05	-0.04	-0.01
	[-1.27]	[5.28]	[5.05]	[1.43]	[-1.35]	[-0.18]
β_{RMW}	0.15	0.07	0.00	0.00	0.19	0.04
	[5.26]	[2.02]	[0.00]	[0.11]	[5.74]	[0.84]
β_{CMA}	0.11	0.09	0.04	0.20	0.23	0.12
	[2.45]	[1.83]	[0.79]	[4.11]	[4.62]	[1.90]
β_{UMD}	-0.01	-0.02	0.06	-0.02	-0.03	-0.02
	[-0.34]	[-1.34]	[3.10]	[-1.31]	[-1.63]	[-1.03]
Panel C: Average number of firms (n) and market capitalization (me)						
n	223	292	329	292	223	
me ($\$10^6$)	1314	1636	808	1455	1384	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the AID strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27	0.27	0.26	0.27	0.22	0.23
			[3.05]	[3.08]	[2.86]	[2.94]	[2.37]	[2.50]
Quintile	NYSE	EW	0.13	0.15	0.11	0.08	0.08	0.05
			[2.02]	[2.34]	[1.82]	[1.25]	[1.20]	[0.82]
Quintile	Name	VW	0.14	0.16	0.14	0.16	0.09	0.11
			[1.75]	[1.90]	[1.64]	[1.91]	[1.05]	[1.34]
Quintile	Cap	VW	0.17	0.17	0.17	0.23	0.14	0.19
			[1.84]	[1.82]	[1.86]	[2.44]	[1.50]	[2.01]
Decile	NYSE	VW	0.36	0.37	0.33	0.34	0.28	0.29
			[3.13]	[3.15]	[2.85]	[2.87]	[2.29]	[2.36]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.23	0.22	0.20	0.20	0.16	0.16
			[2.53]	[2.42]	[2.20]	[2.26]	[1.72]	[1.80]
Quintile	NYSE	EW	-0.08					
			[-1.17]					
Quintile	Name	VW	0.10	0.10	0.08	0.10	0.03	0.05
			[1.20]	[1.26]	[0.98]	[1.14]	[0.41]	[0.58]
Quintile	Cap	VW	0.12	0.11	0.11	0.14	0.08	0.10
			[1.35]	[1.22]	[1.20]	[1.55]	[0.85]	[1.12]
Decile	NYSE	VW	0.31	0.30	0.26	0.27	0.20	0.21
			[2.67]	[2.54]	[2.24]	[2.28]	[1.69]	[1.77]

Table 3: Conditional sort on size and AID

This table presents results for conditional double sorts on size and AID. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on AID. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high AID and short stocks with low AID .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

	AID Quintiles					AID Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.94	0.93	0.98	0.83	0.74	-0.20	-0.20	-0.25	-0.21	-0.30	-0.26
	[3.38]	[3.66]	[3.67]	[3.11]	[2.71]	[-1.98]	[-1.93]	[-2.44]	[-1.99]	[-2.84]	[-2.49]
(2)	0.79	0.82	0.84	0.91	0.98	0.18	0.22	0.14	0.08	-0.01	-0.05
	[2.92]	[3.21]	[3.42]	[3.60]	[3.72]	[1.50]	[1.82]	[1.19]	[0.65]	[-0.09]	[-0.41]
(3)	0.88	0.78	0.89	1.01	0.88	-0.00	-0.01	-0.01	-0.05	-0.01	-0.04
	[3.59]	[3.36]	[3.96]	[4.03]	[3.52]	[-0.01]	[-0.11]	[-0.07]	[-0.42]	[-0.09]	[-0.33]
(4)	0.83	0.76	0.85	0.74	0.85	0.02	0.02	-0.02	0.04	-0.04	0.01
	[3.69]	[3.44]	[4.05]	[3.45]	[3.78]	[0.22]	[0.17]	[-0.17]	[0.38]	[-0.40]	[0.05]
(5)	0.51	0.74	0.52	0.54	0.81	0.30	0.29	0.30	0.29	0.25	0.25
	[2.66]	[3.72]	[2.86]	[2.75]	[4.10]	[2.71]	[2.63]	[2.65]	[2.58]	[2.20]	[2.20]

Panel B: Portfolio average number of firms and market capitalization

	AID Quintiles: Average n					AID Quintiles: Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	139	138	138	139	139	16	12	12	12	15
(2)	42	42	41	42	41	26	26	25	26	26
(3)	32	32	32	32	32	49	49	49	49	49
(4)	29	29	29	29	29	116	115	111	112	115
(5)	31	31	31	31	31	938	1425	888	1270	1067

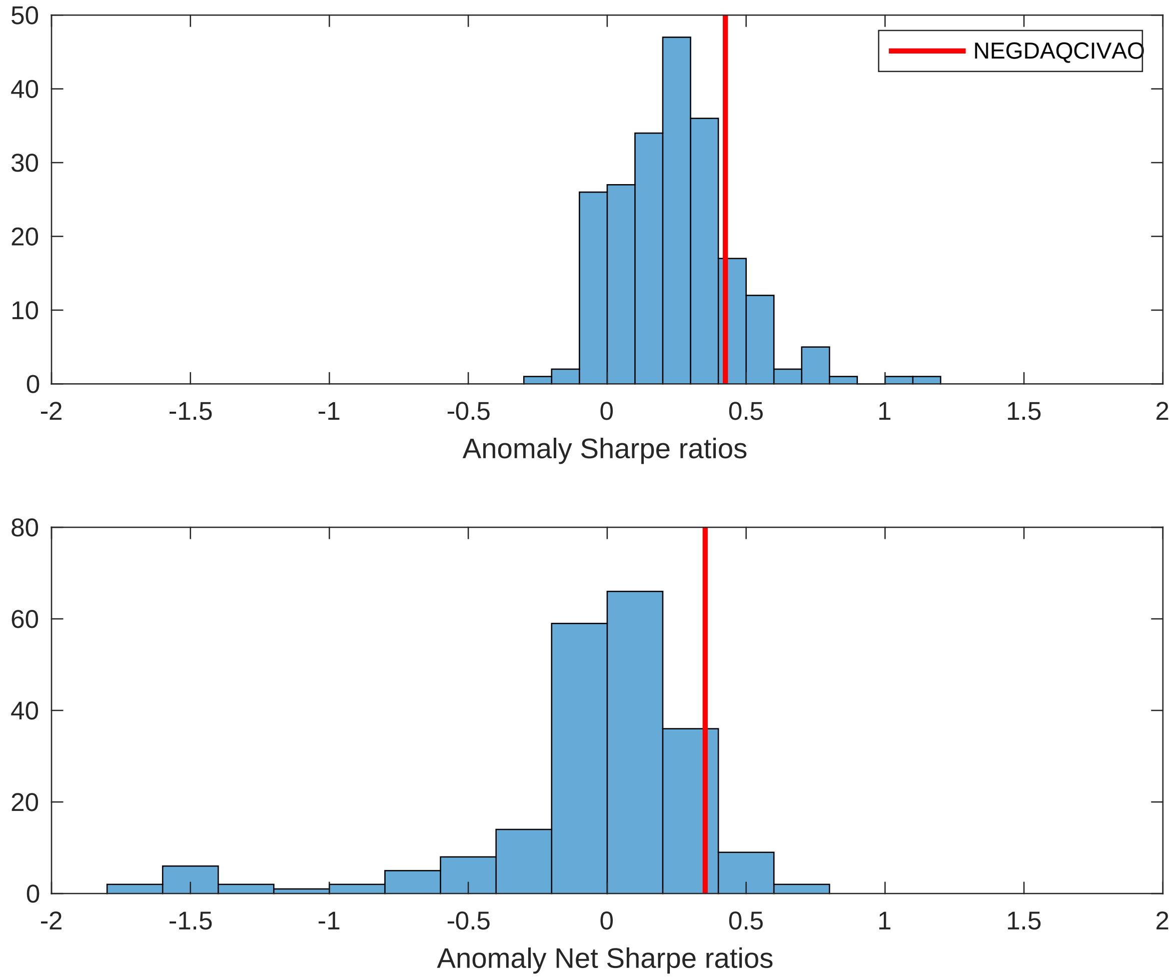


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the AID with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

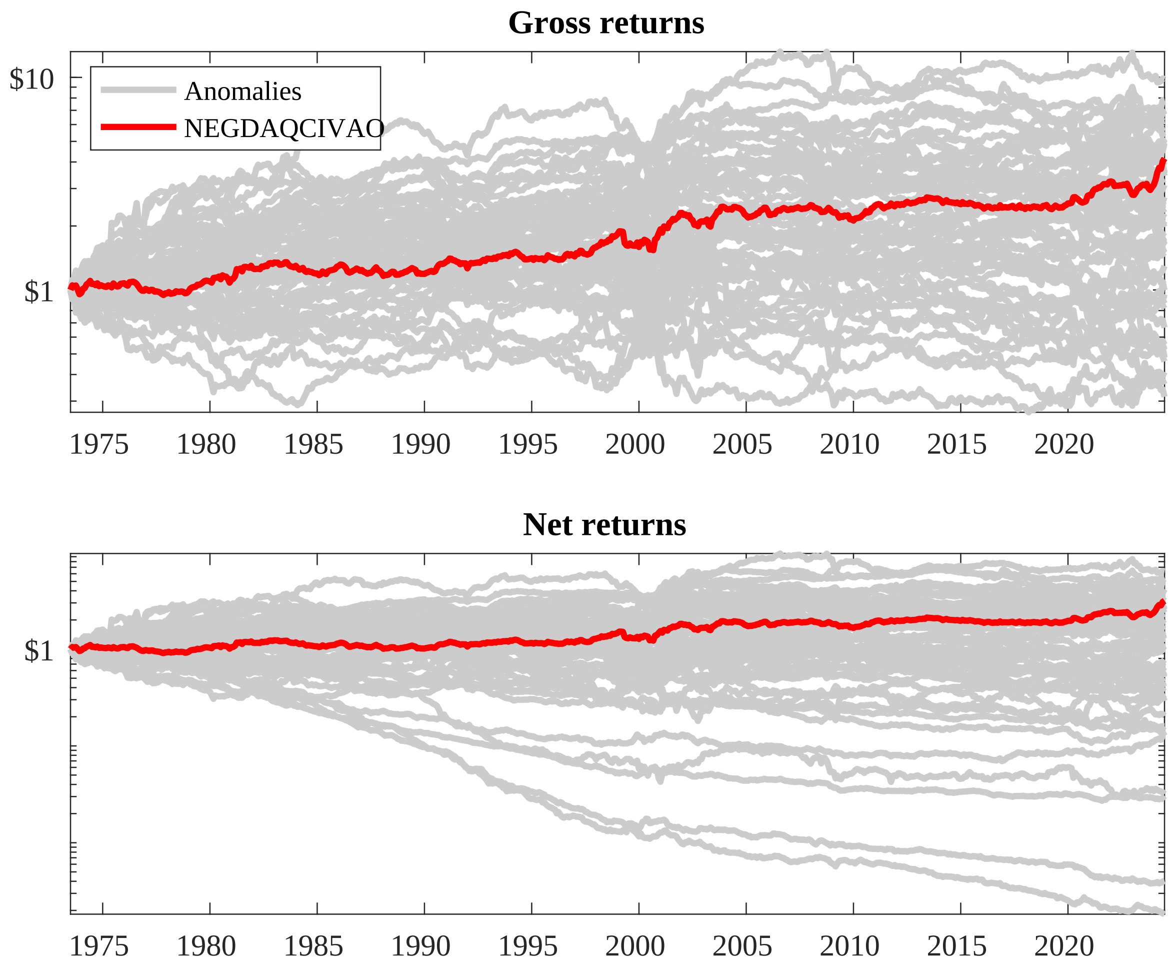
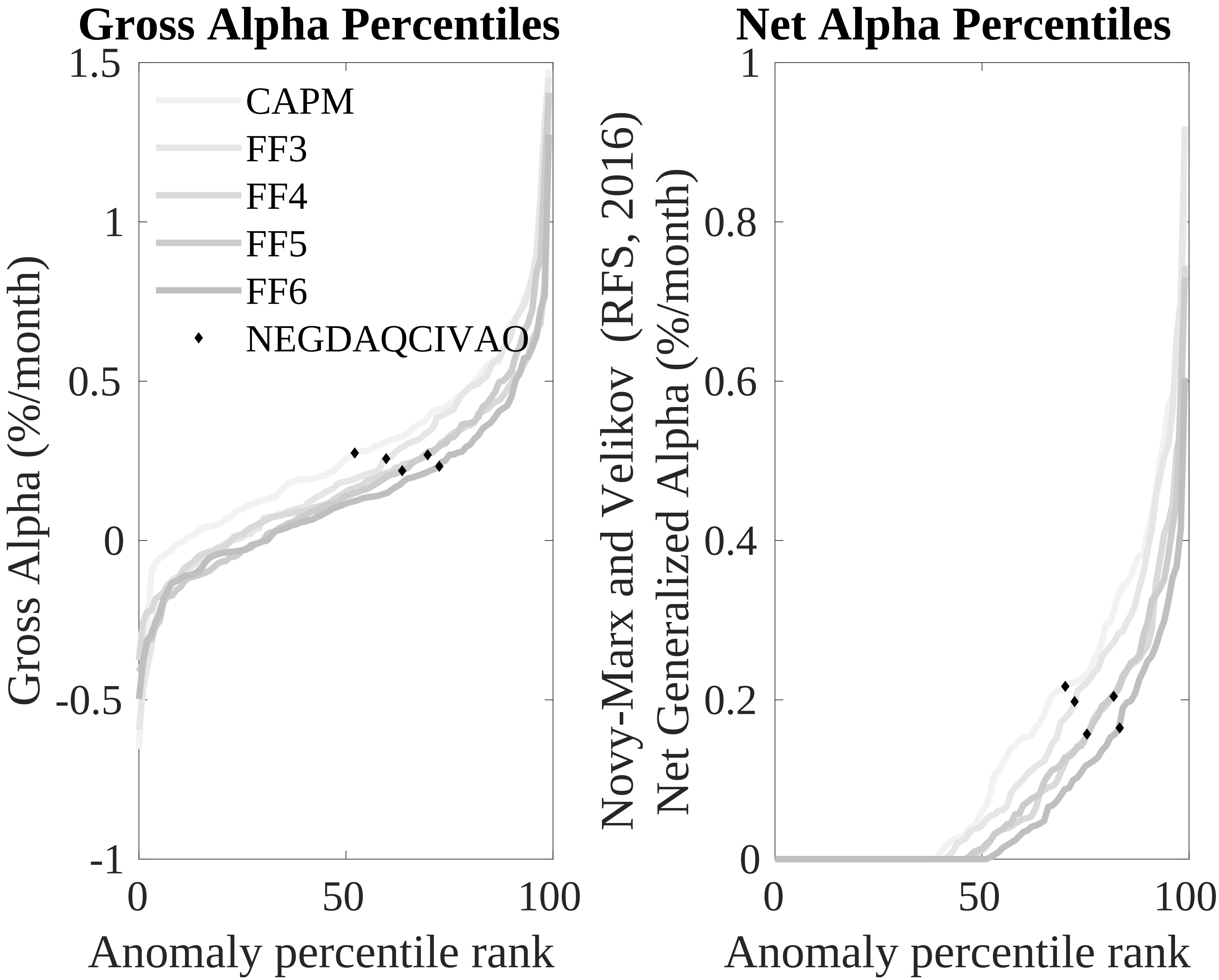


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the AID trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDAQCIVAO
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

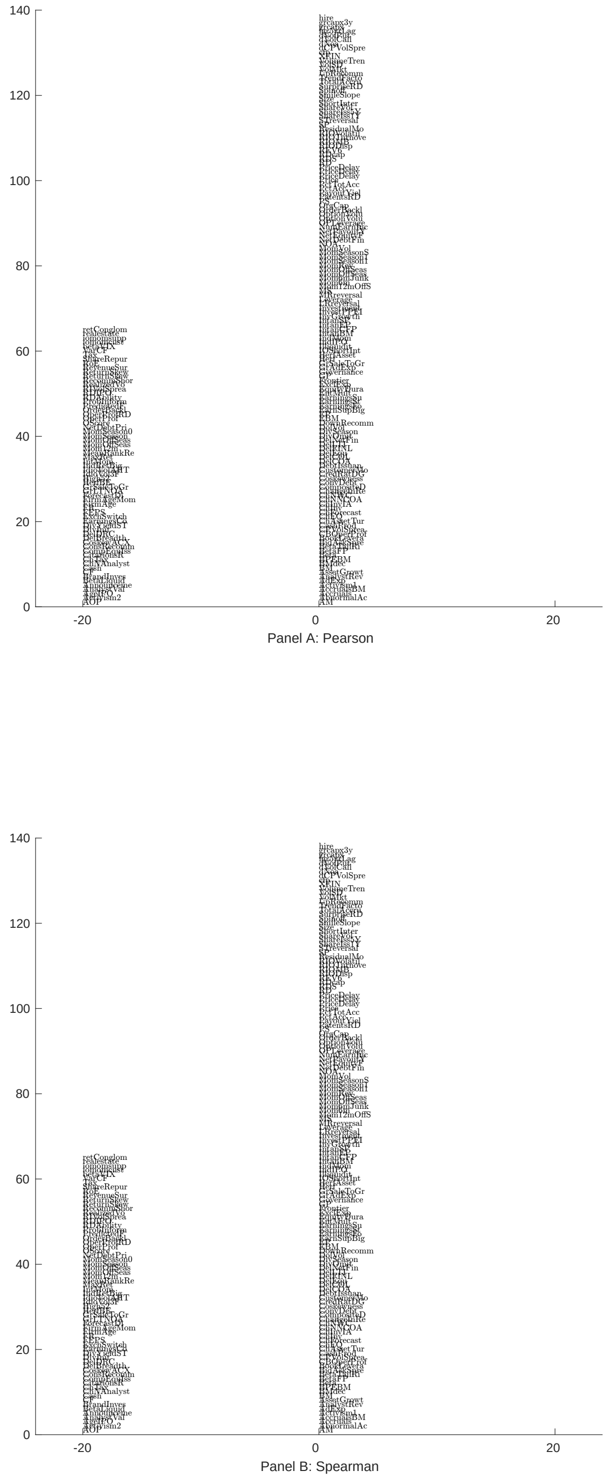


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 203 filtered anomaly signals with AID. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

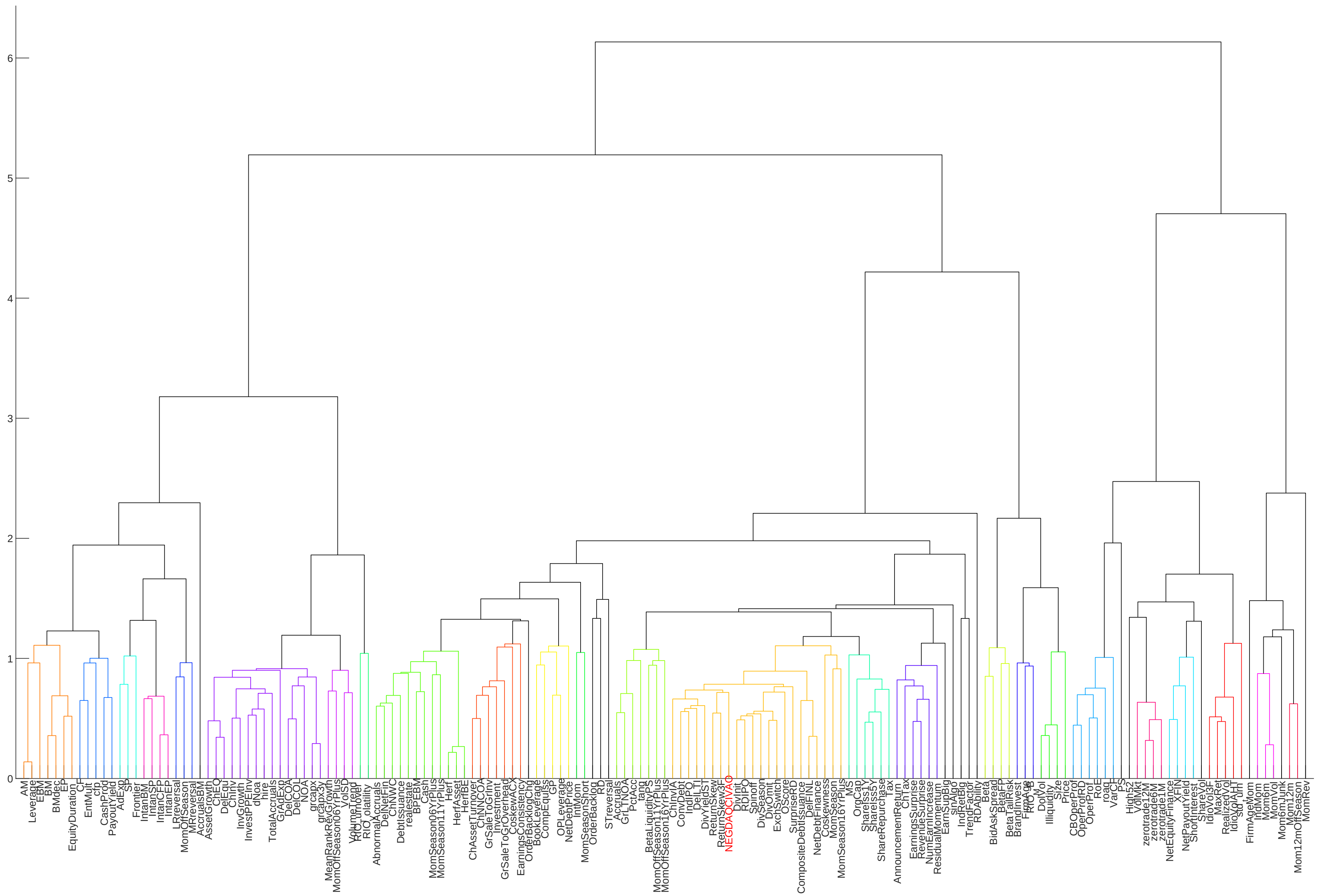


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

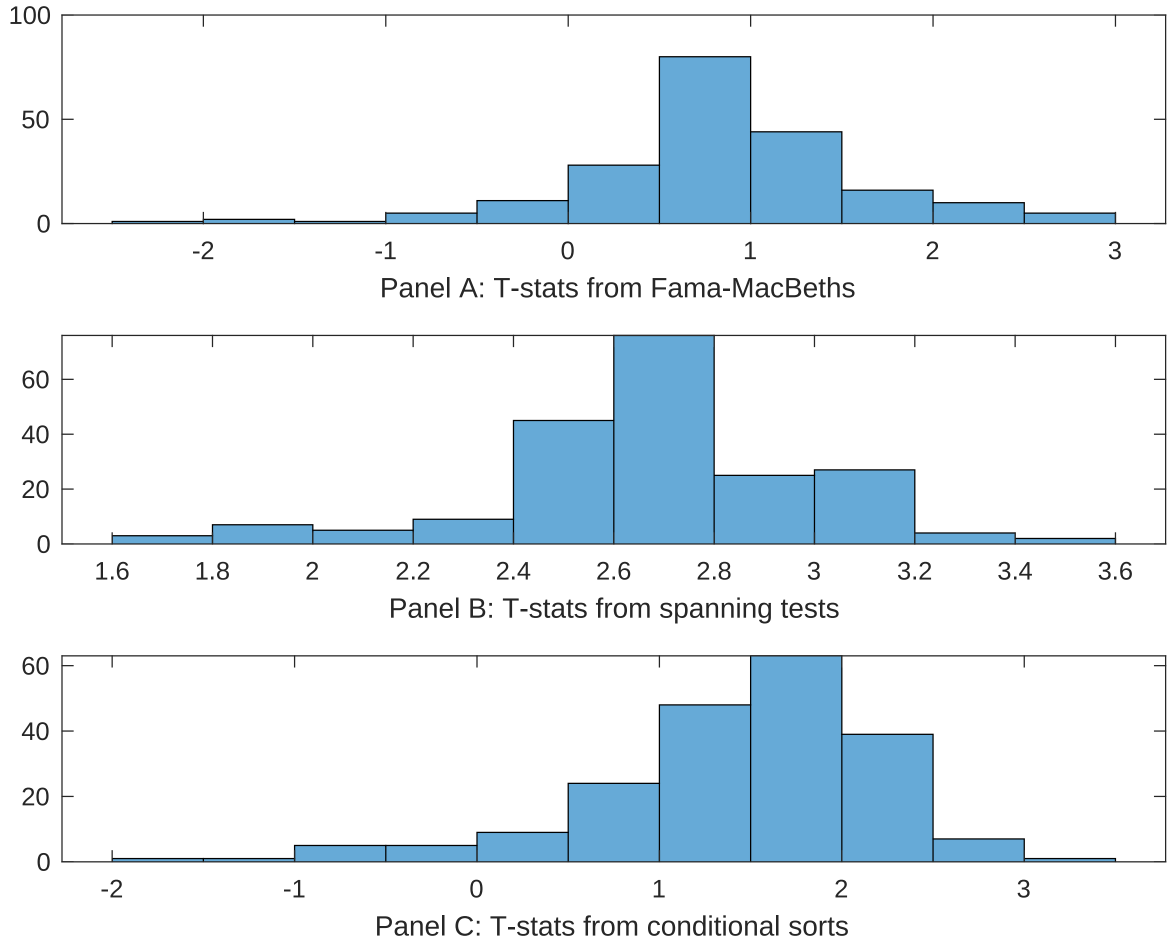


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of AID conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AID} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AID} AID_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AID,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on AID. Stocks are finally grouped into five AID portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted AID trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on AID. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{AID} AID_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Asset growth, Change in net financial assets, Change in equity to assets, Growth in book equity, Change in current operating assets, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14	0.13	0.13	0.18	0.13	0.13	0.12
	[6.14]	[5.60]	[5.71]	[7.00]	[5.66]	[5.81]	[5.26]
AID	-0.66	0.25	0.13	0.84	-0.11	-0.17	-0.67
	[-1.88]	[0.07]	[0.37]	[0.24]	[-0.03]	[-0.47]	[-1.82]
Anomaly 1	0.11						0.74
	[7.96]						[4.65]
Anomaly 2		0.52					0.29
		[2.52]					[0.13]
Anomaly 3			0.16				0.73
			[4.21]				[1.23]
Anomaly 4				0.48			-0.19
				[4.58]			[-1.37]
Anomaly 5					0.17		0.35
					[4.42]		[0.95]
Anomaly 6						0.18	0.93
						[4.99]	[2.19]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the AID trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{AID} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Asset growth, Change in net financial assets, Change in equity to assets, Growth in book equity, Change in current operating assets, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21	0.24	0.21	0.21	0.22	0.20	0.22
	[2.28]	[2.58]	[2.29]	[2.24]	[2.35]	[2.18]	[2.38]
Anomaly 1	26.10						22.96
	[4.91]						[3.53]
Anomaly 2		-12.00					-7.67
		[-2.72]					[-1.51]
Anomaly 3			13.75				3.72
			[3.00]				[0.51]
Anomaly 4				11.90			-5.42
				[2.38]			[-0.73]
Anomaly 5					6.20		-0.74
					[1.36]		[-0.15]
Anomaly 6						9.64	5.23
						[2.07]	[1.08]
mkt	1.32	1.05	1.19	1.70	1.20	2.62	1.68
	[0.63]	[0.49]	[0.56]	[0.79]	[0.56]	[1.17]	[0.76]
smb	-2.24	-1.02	-0.29	-0.70	1.00	2.76	-0.75
	[-0.70]	[-0.32]	[-0.09]	[-0.22]	[0.30]	[0.77]	[-0.20]
hml	-2.07	-0.54	-2.65	-2.17	-3.34	0.28	-0.75
	[-0.51]	[-0.13]	[-0.64]	[-0.52]	[-0.73]	[0.07]	[-0.17]
rmw	4.22	2.69	5.36	4.52	4.84	-1.63	0.23
	[1.02]	[0.64]	[1.28]	[1.08]	[1.14]	[-0.32]	[0.04]
cma	-19.02	9.40	-1.18	1.12	9.35	6.22	-18.74
	[-2.14]	[1.48]	[-0.15]	[0.14]	[1.40]	[0.89]	[-2.01]
umd	-1.68	-1.98	-2.09	-2.51	-1.96	-2.27	-1.50
	[-0.80]	[-0.93]	[-0.99]	[-1.18]	[-0.92]	[-1.06]	[-0.71]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	4	2	2	1	1	1	4

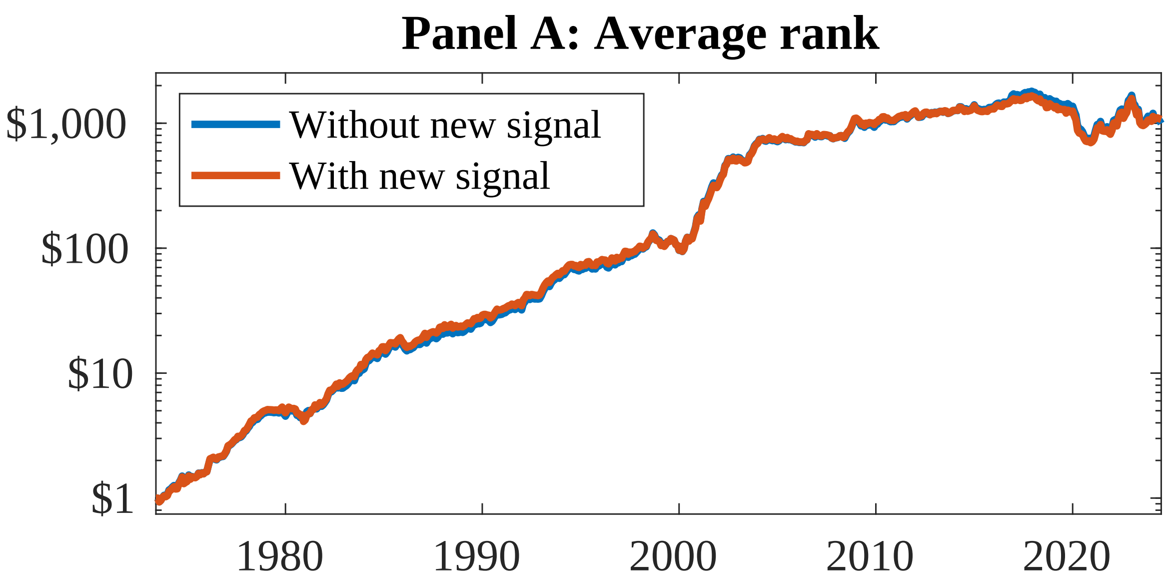


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as AID. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.

Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.

Berk, J. B., Green, R. C., and Naik, V. (1999). Optimal investment, growth options, and security returns. *Journal of Finance*, 54(5):1553–1607.

Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2008). Dissecting anomalies. *Journal of Finance*, 63(4):1653–1678.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.

Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.

Lettau, M. and Ludvigson, S. (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Rietz, T. A. (1988). The equity risk premium a solution. *Journal of Monetary Economics*, 22(1):117–131.

Tsai, J. and Wachter, J. A. (2016). Disaster risk and its implications for asset pricing. *Annual Review of Financial Economics*, 8:219–252.

Wachter, J. A. (2006). A consumption-based model of the term structure of interest rates. *Journal of Financial Economics*, 79(2):365–399.

Wachter, J. A. (2013). Can time-varying risk of rare disasters explain aggregate stock market volatility? *Journal of Finance*, 68(3):987–1035.

Yogo, M. (2006). A consumption-based explanation of expected stock returns. *Journal of Finance*, 61(2):539–580.

Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.